

Senior Autonomous Control Systems Engineer

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"Driven by passion, innovation, and a relentless pursuit of progress, I strive to create solutions that redefine possibilities in embedded, IoT and Robotics"

SUMMARY

I am a **Robotics Engineer** with over **15 years of experience** specializing in both software and hardware development for **embedded systems, robotics, and real-time communication devices**. I have extensive experience end to end product development, programming embedded systems, developing **robotics applications**, and working with **front-end software** using **Python, C++, and ROS** (Robotics Operating System). Throughout my career, I've contributed to various projects with an emphasis on integrating hardware and software systems to create innovative, efficient solutions.

I am a hands-on person and am looking for roles that allow me to expand the horizons of my knowledge. My motivation lies in the products and challenges companies aim to solve. I enjoy bringing ideas to life, managing projects from concept to production, and developing solutions that significantly improve efficiency and solve complex problems. I take pride in leading product development and playing a key role in creating groundbreaking robotics technologies.

PROFESSIONAL SKILLS

- **Leadership & Team Management:** Proven ability to lead cross-functional teams, ensuring collaboration, productivity, and high-quality results.
- **Stakeholder Management:** Skilled in managing relationships with key stakeholders, aligning project goals, and ensuring customer satisfaction throughout the development process.
- **Technical Marketing:** Able to translate complex technical concepts into clear, marketable messaging, and effectively communicate product value propositions to both technical and non-technical stakeholders.
- **Documentation, Technical Writing & Presentation:** Demonstrated expertise in technical writing and delivering presentations, with published papers and successful technical talks at conferences.
- **End-to-End Product Development:** Led multiple teams through the complete product development lifecycle, including **mechanical design, electronics design, prototyping, 3D printing, and digital manufacturing techniques**. Focused on optimizing workflows to enhance efficiency and reduce costs.
- **Workflow Optimization:** Passionate about optimizing production cycles and workflows to improve overall efficiency and reduce operational costs.

PERSONAL SKILLS

- Excellent **communication skills** in English, both written and verbal.
- **Motivates and guides cross-functional teams** to deliver high-quality results.
- **Kind and Supportive:** Always approachable, empathetic, and willing to lend a helping hand, creating a positive and inclusive work environment.
- Strong **interpersonal skills** that enable effective collaboration and communication across teams.
- **Quick Learner & Adaptability:** Fast to adapt to new technologies and methodologies, continually staying ahead.
- **Adaptable and curious**, always eager to learn and take on new challenges.
- **Cooperative and flexible**, able to thrive in diverse work environments.
- Highly **responsible**, ensuring goals and deadlines are met with precision.
- **Ambivert**, balances independent work with engaging collaboration.
- **Ambitious**, driven to continuously improve and achieve goals.
- **Efficient and systematic**, approaching tasks with a methodical mindset.
- **Dynamic and energetic**, bringing enthusiasm to every project.
- **Humorous**, contributing to a positive and motivating atmosphere in the workplace.
- **Problem-Solving:** Strong analytical skills to identify and resolve issues effectively, ensuring smooth project execution and product development.
- **Mentoring:** Passionate about mentoring junior team members, fostering their growth, and sharing knowledge to strengthen team capabilities.

KEY TECHNOLOGIES

ROS; ROS2; Gazebo; RVIZ; Plot juggler; Fusion360; Eagle; Rapid prototyping; Linux; GIT; AWS (prior experience); CAN; UART; RS232; RTOS; I2C; SPI; C++; C; Python; JavaScript; Plus One GUIDE; Plus One Service Tool; MQTT;

NB-IoT; IoT; IoET ; DFM; Manufacturing; 3D printing; WebSocket; Angular; React; JS; HTML; CSS; Robotics Data Visualisation; HIL; SIL; AMR; SLAM; Navigation stack; Planners; Drones; Mission planners; Simulation.

EMPLOYMENT

Lead Robotics Engineer	The Mobile Robot Company (Denmark)	June 2025 – present
- Led platform engineering initiatives for autonomous mobile robot (AMR) systems, developing internal tooling, infrastructure, and engineering workflows to improve developer productivity, deployment readiness, and system reliability. - Designed and implemented DevOps practices and CI/CD pipelines utilizing cloud-based container builds, dependency caching, and incremental compilation strategies to accelerate development cycles and reduce build times. - Developed Software-in-the-Loop (SIL) and Hardware-in-the-Loop (HIL) validation frameworks, reducing reliance on physical robots during development and enabling earlier detection of integration issues. - Developed high-fidelity robot simulations replicating real-world kinematics, motion behaviour, and operational characteristics, maximizing robot availability for demonstrations, testing, and customer deployments. - Created Safety PLC and battery system emulators with interactive visual interfaces, enabling software validation and integration testing without physical hardware while accurately emulating real-world device communication. - Architected a modular robotics software platform with clearly separated driver, simulation, logic, navigation, and shared-service layers, improving maintainability, scalability, and parallel development across the engineering team. - Developed an automated safety-zone generation tool using robot URDF models to calculate dynamic safety fields for safety LiDAR systems based on speed and steering-angle-dependent operation, reducing configuration effort from months to hours. - Drove engineering process improvements, developer enablement initiatives, and deployment readiness activities, proactively mitigating risks and improving delivery speed across the robotics development lifecycle.		
Sr. Autonomous Control Systems Engineer	Danfoss Power Solutions (Denmark)	July 2023 – April 2025
<i>Autonomous Baggage Handler - Robotics</i> - Assisted the team in Sensor Fusion, Hardware testing and Data Visualisation. - Developed Plus one GUIDE(Danfoss) compliance blocks Sensata Radar (Preview Sentry). - Evaluated Sensata Radar (Sentry 79) for its accuracy compared other radars. <i>Autonomous Systems Dashboard – Robotics Visualisation</i> - Evaluated visualisation using RVIZ for Hardware in Loop System by developing POC. - Evaluated Plus one Service tool based visualiser using its internal Java Script functionality. - Developed Libraries for Plus One Service tool using WebSockets to make design of web visualiser modular. - Architected and designed a web-based plugin to visualise Ouster Lidar point cloud. - Developed POC for web visualiser for real-time data from robot controller and sensors allowing for 30% improvement in efficiency in developing, testing, and tuning of robot application on Plus one GUIDE(Danfoss).		
Lead Robotics Engineer	Sinergia Media Labs (India)	Dec 2021 -April 2023
<i>Fill and Finish - Robotics</i> - Designed and architected a standalone robotic cell for vial filling and capping in ISO5 aseptic conditions. - Designed using two ‘Staubli TX260’ aseptic manipulators to fill 500-600 vials an hour. - Sped up development process with regular course corrections and mentoring sessions for the team. - Designed an optimised hybrid gripper using ‘Schunk EGL gripper’ which can do mechanical & pneumatic tasks. - Simulated the system in ROS2 Gazebo. <i>Smart Dose - Robotics</i> - Architected and built robotic cells for flexible automating assembly of part of the assembly line for vaccine delivery. - Designed using three ‘Omron TM5-900’ manipulators to complete assembly in under 2 minutes. - Designed multiple grippers using ‘Schunk EGH 80’ to make the processes mechanically reliable and repeatable.		
Head Research and Development	Robo Inventions (India)	Sept 2010 – Nov 2021
<i>NavApp (2021) – Robotics Visualisation</i> - Designed and developed a Human Machine Interface for AMR robot. - Designed scalable to be incorporated directly into fleet management software. - Lead the UI/UX team in designing an intuitive web app for visualising data as well as sending commands to robot. - Derived and improved on the ROS2DJS and NAV2DJS libraries to implement new functions for LIDAR data plotting, Path plotting, No-Go zones creation and waypoint navigation. - Developed custom Java Script Lib for visualisation of robot data – Map, Battery %, ETA for current task. - Created a front-end task sequencer for sending tasks to the robot. - Improved UI functionality of front-end Map – Zoom, Pan, and interactable markers for robot path and POI.		

Autonomous Robotic Carrier (2020) - Robotics

- Designed and manufactured production ready differential drive AMR with 200kgs payload capacity.
- Wrote the Hardware Abstraction Layer for sensors and motors.
- Architected and designed the software stack in ROS Noetic.
- Integrated and tuned with various SLAM and Planning algorithms.
- Lead the development of a unique compliant mechanical design in Fusion360.
- Developed embedded hardware and firmware for kinematics and servo motor control.
- Lead cross team collaborations for product design and engineering teams.
- Implemented robot perception using RP-lidar and ROS sensor fusion.
- Tested and validated hardware and software.

Humanoid Intended for Robotics Operation (2018) - Robotics

- Designed and built Humanoid robot for customer interactions.
- Architected and designed the software stack in ROS Kinetic.
- Wrote the Hardware Abstraction Layer for sensors and motors.
- Designed mecanum drive system from self-developed mecanum wheels and industrial servos.
- Lead a design and manufacturing team to build a unique body from composite materials.
- Derived and improved a unique & economical manufacturing process for CAD to composite conversion.
- Developed embedded hardware and firmware for motor control, kinematics and reading sensor data.
- Integrated ROS based locomotion stack and joystick-based teleoperation node for Holonomic robots.
- Developed ROS based UI to display faces with emotions on the robot's screen.

Electro-Pneumatic Positioner valve (2019) - Embedded

- Architected and developed a positioner valve and internal electro pneumatic components including the control board and micro-flapper valve.
- Acted as consultant for external development vendor as a solution architect.
- Mitigated bottlenecks and development issues during Electronics, firmware and mechanical development phases.
- Facilitated the process for the customer to move the product into production.

Modular Prosthetic Wrist (2019) - Product development

- Designed and developed a low-cost modular prosthetic wrist for social hardware company using Atmega328 and electronics muscle sensor in collaboration with Fusion360.
- Board consisted of power management, sensing and motor control.
- Designed the product in Fusion360 to be 3D printable in flexible materials and using generative design.

Brake pad Testing Dataloggers (2016-2022) - Embedded

- Developed multiple remote Data loggers for the RnD division of automotive parts manufacturer (TVS) for testing of asbestos free compositions for Brake pads of trucks and performance vehicles.
- Each individual datalogger would collect one of the various parameters – Wheel RPM, Brake pad Temp, Steering angle, Gear position, etc. This would allow the customer to create a log of the different driving profiles.
- Designed a common mother board based on ESP32, lithium battery charging and protection circuit and low power voltage regulator.
- Designed Bluetooth Serial based protocol to mediate communication between the remote datalogger and the customers interface device.
- Multiple loggers would be connected to single interface device allowing for monitoring of all the params inside the vehicle cab without lot of wiring and allowing for quick and modular fitting of the sensors.

Drone Control Board

- Designed an avionics board for multirotor based on the Arduino mega and running on the Ardupilot software.
- Integrated the 6DOF sensor MPU6050, 3DOF magnetometer and BMP280 I2C barometer to create an IMU.
- Integrated 950Mhz long range Zigbee for telemetry and ground control.
- Remote control was achieved with the use of a 2.4Ghz radio.

Electrobricks (2015) – STEM kit

- Designed and developed and DIY STEM kit allowing which helps explain complex scientific concepts in tangible manner.
- Architected the Kit to be colour coded according to different groups to make it easier to understand.
- Made the kit to be modular and pluggable, to allow for infinite combination.
- Designed the Kit using only analog electronics.
- Marketed the product to the Govt. of Kerala and were able to implement the 'Electronics at School' project state wide encompassing more than 3000 school.
- Conducted teacher training for 1200 staff of school from all over the state.
- Applied Design for Manufacturing (DFM) techniques to produce 3700 kits indigenously in India in a short period.

Q (2019) – STEM kit

- Developed an improved version of the Electrobricks kit with IoT connectivity.

- Designed the kit to be able to connect to a E learning app to allow for HW/SW interaction.
- Designed HW connector influenced from the MagSafe connector to make it easier for smaller hands.
- Developed a Protocol layer on the Serial communication layer to allow bricks to process info and communicate to the main controller.

Nava Technologies - automatic coconut palm tapping robot

- Solved technology challenges in developing the automatic coconut palm tapping robot through consultation.
- Applied knowledge of maple tree and coconut palm tapping to improve prototype efficiency.
- Contributed to the initial prototype, addressing key technical issues.
- Worked with teams to integrate solutions into the coconut palm tapping system.
- Provided expertise in sensor integration and automation for palm tapping.

Quasi Direct Drive Actuator for Robotics application – Embedded

- Developed a BLDC Servo motor with integrated driver to allow for high torque and reduces complexity AMR robot.
- Designed system with ESP32 microcontroller, heavily influenced by the Moteus Controller (Open source)
- Evaluated concept of Sensor-less feedback to reduce system complexity

Thing Z Platform - IOT

- Designed modular IOT mother board using ESP32 to communicate to a clients AWS cloud platform using MQTT
- Developed firmware to be scalable for multiple products of the customer.
- Architected system design for HW, SW and UI (App).
- App included the feature to configure the device via Wifi.

N-Lock Notification Device - IOT

- Designed and developed a IOT device for notification when door is opened during non-working hours.
- Designed Ultra-low power IOT board using ESP32 and U-blocks NB-IoT controller working on the MQTT protocol.
- Designed and developed Ultra-low power power-supply and firmware to allow device to run 2 years with one cell.
- Designed an app to configure the device via Bluetooth.

SoftSwitch (2018) – IOT

- Designed and developed an IOT HW eco-system based on MQTT protocol with switches, regulators, sensors and bells compatible with Home Assistant.
- Connected Home Assistant to Amazon Alexa for Voice control.
- Designed circuit board based on ESP32 and solid-state hardware (BT139) with following functionality – Touch, Lighting for the touch buttons, drivers for AC switching plus dimming and silent functioning.

InduinoX (2011) – Open-Source Microcontroller board

- Developed microcontroller training board based on the Arduino using Atmega 328 and CH430 USB to UART chip.
- Added IR communication (TSOP0038), RGB Led, LDR and RTC.
- Developed training modules for key concepts – UART, IR (RC5) and I2C communication, RTC, Colour mixing etc.

RIduino (2010) – Open-Source Microcontroller board

- Developed indigenous Arduino based board with Atmega 328 and CP2102 low-cost USB to UART controller.
- Developed daughter board with LCD and button for developing games and remote controllers.
- Developed training modules with key concepts – LCD, shift registers (74HC595), Led matrix displays, multiplexing.
- Developed robotics daughter board with motor driver L298 and L293.
- Developed robotics training modules with key concepts - motor driving, sensors (IR and Sonar), Control algo. (PID).

EDUCATION

B-Tech in Electronics & Communication	Pondicherry University	July 2005 – June 2009
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LANGUAGES

Read . Write . Speak	Speak	Basic understanding
English, Hindi.	Malayalam, Tamil.	Dansk (started learning).

PUBLICATION

- Co-authored (2022): IEEE Paper on ‘*A Study on Intuitive Interaction between Mobile Robots and Humans*’.

AWARDS AND ACCOLADES

- **40 under 40 (2018):** Awarded by ‘The New Indian Express’ (Newspaper) for developing Robots in India.

- **STEM (2015 onward):** Spearheaded the **STEM movement** along with '*Kerala IT Mission*' and started the '*Electronics @ School*' project being implement in over **3000 schools** in Kerala.
- **Guest lectures: (Dec 2019)** - Delivered at the '*University of Tasmania*' on '*Robotics and Entrepreneurship*' during the Strategic Entrepreneurship program & **(Dec 2020)**- Delivered at the '*Chitkara University*' on '*EDGE computing and IOT*'.

VOLUNTEERING ACTIVITIES

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- **Mentor (Nov 2024 Onwards)** – Mentored SDU Sønderborg, Robotics club students in developing Micromouse Robot.
 - **Consultant (2015 onwards)** – Ed. Tech. Consultant to 'Kerala IT Mission' for developing and improving course curriculum for STEM courses provided in over 3000 Govt. schools in the state.
 - **STEM Trainer and Mentor (2005 – 2009)** – Setup a Robotics club in school during my final years of schooling and I trained and mentored many my juniors in school about Robotics and Embedded systems.

HOBBIES

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- **Electronics & Robotics** – Inventing interesting products to make my life simpler.
 - **Traveling with the family**– Love traveling with my wife and 3-year-old to explore new places and cuisines.
 - **Radio Controlled Aircrafts** – Build and fly model aircrafts with the family to make new friends and connect with old ones.

REFERENCES

References are available upon request.